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Predicting and Visualizing Short-Term House Price Changes in England and Wales using AI

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I confirm I have proof-read, and quality checked the report, ensuring it is free from grammar, spelling, and formatting errors, and meets high standards of clarity, coherence, and presentation.

MSc FPR Declaration

This report is submitted in partial fulfilment of the requirement for the degree of: Master of Science in Data Science and Analytics, at the University of Hertfordshire (UH).

I hereby declare that the work presented in this project and report is entirely my own, except where explicitly stated otherwise. All sources of information and ideas, whether quoted directly or paraphrased, have been properly referenced in accordance with academic standards. I understand that any failure to properly acknowledge the work of others could constitute plagiarism and may result in academic penalties.

I did not use human participants in my MSc Project.

I hereby withhold permission for the report to be made available on the university website provided the source is acknowledged.

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1. Abstract

Predicting house prices accurately, particularly in dynamic markets like London, poses significant challenges for traditional modelling techniques because of the complexity and non-linearity of the factors involved. The aim of this project is to investigate to what extent can machine learning and deep learning models improve the short-term prediction accuracy for houses in England and Wales (London in particular) compared to baseline methods with combined HM Land Registry Price Paid data (PPD) and Energy Performance Certificate (EPC) data. The methodology involved merging and preprocessing the PPD and EPC datasets, addressing memory constraints and high-cardinality feature challenges by employing label encoding. Five models were built and compared to one another including Linear Regression (LR), Random Forest (RF), XGBoost (XGB), Multi-Layer Perceptron (MLP), and Long Short-Term Memory (LSTM). Performance was evaluated in detail using MSE, MAE, RMSE, and R^2 by training the model on data excluding 2024 data and then evaluating final hold-out set from 2024. Furthermore, XGBoost appeared to excel as it was more accurate than the others, boasting an R^2 of approximately 0.82 on the test set and an impressive 0.83 on the 2024 hold-out data, which all compared to a dismal performance set by the LR baseline ($R^2 \approx 0.55$ Test, 0.59 Hold-Out). RF ($R^2 \approx 0.78$ Test, 0.77 Hold-Out) and MLP ($R^2 \approx 0.75$ Test, 0.74 Hold-Out) also significantly improved upon LR results, while the implemented LSTM model ($R^2 \approx 0.63$ Test, 0.64 Hold-Out) reportedly proved less effective for that specific application. Features of great importance included location, size of property (from EPC), and temporal features, as evidenced by feature importance analysis. Thus, it can be concluded that XGBoost performs very well for real estate price prediction application and has superior predictive power over traditional and simple deep learning approaches. Accessing this EPC data proved excellent for valid findings. Future work can be invested in alternative encoding methods (target encoding, for instance), hyperparameter tuning, and more specialized deep learning architectures.

2. Introduction

The project falls into the field of Data Science and Analytics, with a focus on using AI for predictive modelling and interactive visualization in the real estate market. Further, it aims to investigate the extent to which the performance of house price prediction models may be improved in England and Wales but limited to London using conventional regression and machine or deep learning models.

2.1. Problem Overview

The real estate market in the UK, especially in London, forms an important part of the national economy of UK and individual wealth. Therefore, it becomes important to predict house prices accurately to serve homebuyers, sellers and investors. The housing market is very complex. Factors influencing housing prices can include economic indicators like interest rates, inflation, or employment and government policies etc. Locality based characteristics such as school quality, crime rates, and amenities can also determine the price of property. Further, house

specific features such as size, number of rooms, general condition, and energy efficiency can also affect house prices.

Furthermore, during uncertain economic times, prediction of buying/selling scenarios is a difficult task for property owners, developers, and investors alike. Conventional methods of forecasting house prices, which mostly contain basic techniques like linear regression models, have proven to be far too simple given the nature of dynamics and non-linear patterns are present in far more complex real estate datasets which can lead to poor results (Rosen, 1974).

Due to advances made in the field of Deep Learning (DL) and Machine Learning (ML), a set of highly advanced models that may work on high-dimensional data and intricate patterns with much higher accuracy are now becoming accessible. (Gu et al., 2020). To this end, the project revolves around comparing the performance of conventional prediction methods like linear regression and random forest with advanced machine learning prediction methods like XGBoost, ANN (Artificial Neural Networks) and LSTM (Long Short-Term Memory) and creating an AI-based visualization tool for visualizing predicted short-term changes in house prices in London. Ethical considerations mainly relating to data privacy (considerably mitigated using public data in this case) and possible biases in the model are quite relevant as well.

2.2. Current Issues

Apparently, some web-based property valuation tools (like Zoopla, Rightmove) are said to be inaccurate, especially for short-term predictions and where markets are highly dynamic (tirboki, 2017). Simplistic models or unavailability of all-inclusive, recent data could be possible reasons for this. Some academic research has been done relating to testing a variety of ML techniques such as Random Forests and Gradient Boosting (Antipov & Pokryshevskaya, 2012; Čeh et al., 2018) which have been shown to yield better results than traditional methods. Deep learning models have been implemented and are noted for being able to capture the complex spatial and temporal dependencies in the data (Mullainathan & Spiess, 2017).

However, there remain quite significant gaps. To begin with, direct comparison of ML algorithms (like XGBoost), with deep learning algorithms (MLP, LSTM), specifically for London house price prediction on integrated Price Paid Data (PPD) (HM Land Registry, 2025) and Energy Performance Certificate (EPC) data (Ministry of Housing, Communities & Local Government, 2025) is quite limited. Price paid data provides transactional data of houses sold in the past beginning from 1 January 1995 while EPC data provides property characteristics (floor area, energy rating, habitable rooms) largely absent in PPD alone. Preliminary exploration of the project determined that PPD alone was not rich enough in features for prediction, hence stressing the need for merging other data. Secondly, the size of UK PPD and EPC data poses enormous issues due to RAM and processing requirements which lead to the same being modestly researched in smaller studies. Lastly, the other thing that needs work is the creation of practical applications that convert output of the model into actionable insight for the end user. The project fills these gaps by systematically comparing regression, ML and DL models on a large integrated PPD and EPC dataset filtered specifically for freehold houses in London, as well as solving the challenges of data processing for an easy-to-use visualisation interface.

2.3. Project Details

The project falls in the domain of Data Science and Analytics with a focus on predicting house prices in London. The main aim of this project is to study and compare the performance of conventional algorithms with machine learning algorithms and deep learning algorithms. It uses a merged dataset of HM Land Registry's Price Paid Data (PPD) (HM Land Registry, 2025) and Energy Performance Certificate (EPC) data (Ministry of Housing, Communities & Local Government, 2025), filtered for freehold house transactions. The key features extracted from EPC data included property attributes such as the floor area, habitable rooms, energy rating, property type etc while PPD data contained features like transaction price and address features (postcode, district, etc.).

A separate csv file was created for the filtered data of London titled london_merged_df_filtered.csv. A csv file was also created from this filter data to further filter it from 2015 to 2024 titled london_merged_df_filtered_2015.csv. Each algorithm was then run on each of the csv files and for label encoding and one hot encoding to collect results and compare them. Moreover, each model was trained and tested for all data except 2024 and then data from 2024 was fed to the model to determine how well each model performs for unseen future data. This means that data other than that from 2024 would be the training and testing set, whereas 2024 data would be held out for final evaluation. The core technical work consisted of data collection, cleaning and preprocessing the data, feature engineering i.e extracting temporal features from price paid dates and encoding categorical features through one-hot encoding or label encoding (after facing dimensionality issues), model implementations (Linear Regression, Random Forest, XGBoost, MLP, and LSTM) and an intensive evaluation of regression metrics (MAE, MSE, RMSE, R^2), and creation of interactive visualisation using Gradio. The project management adopted the iterative approach and was allowed to respond flexibly to the challenges that emerged during data processing and experimentation with the models.

2.4. Aims and Objectives

The primary aim of this project is to evaluate and compare the performance of selected conventional algorithms with machine learning models and deep learning models in the prediction of freehold house prices using integrated PPD and EPC data in London. To achieve this aim, the following objectives were set:

2.4.1 Objective 1: Literature Review and Algorithm Finalization

Target Date: February 21, 2025

1. **Task Description**

I conducted the literature review on the project as stated in the report on real estate prediction and finalized algorithms to be selected. This includes an analysis of important algorithms for house price prediction and some existing visualization tools (such as Zoopla's House Price estimator).

2. **Deliverables**

- Literature review as in the section below.
- A list of chosen regression algorithms (Linear Regression, Random Forest) and deep learning algorithms (ANN, LSTM).

2.4.2 Objective 2: Collection, preprocessing, and cleaning of collected data

Target Date: February 28, 2025

1. **Task Description**

Collection, preprocessing, and cleaning of real estate market data has been done.

2. **Deliverables**

- Clean and processed dataset.
- Python scripts (in a Jupyter Notebook) on the actual data-cleaning process.

2.4.3 Objective 3: Training for the Baseline ML Models

Target Date: March 14, 2025

1. **Task Description**

The aim here is to implement baseline predictive models using linear regression and random forest algorithms. Then, train these models on the pre-processed dataset and report their performance metrics. The same has been done.

2. **Deliverables**

- Source code and scripts using baseline models.
- Performance metrics.

2.4.4 Objective 4: Development of Deep Learning Models

Target Date: March 28, 2025

1. **Task Description**

The goal here is to implement deep learning models using ANN and LSTM algorithms. Then, train these models on the pre-processed dataset and report their performance metrics. The same has been done.

2. **Deliverables**

- Source code and scripts using deep learning models.
- Performance metrics.

2.4.5 Objective 5: Building the Interactive Visualization Tool

Target Date: April 11, 2025

1. **Task Description**

Create an interactive dashboard to visualize past data and model predictions. The same has been done.

2. **Deliverables**

- A working interactive visualization tool implemented in Python.

2.4.6 Objective 6: Evaluation of Models Against Literature Review

Target Date: April 18, 2025

1. **Task Description**

Evaluation of all models in their developed form, where their performance will be compared against insights discovered and benchmarked through the literature review. The same is as in the report below.

2. **Deliverables**

- Results of the comparison between primary and secondary research.
- Comparative charts and tables highlighting the strengths and weaknesses across models.

2.4.7 Objective 7: Write-up the draft Final Project Report

Target Date: April 22, 2025

1. **Task Description**

The project preparation will continue with the detailed documentation, including methodology, experimental results, model evaluations and a concise discussion of ethical, legal, professional and societal implications.

2. **Deliverables**

- Draft version of the Final Project Report.

2.4.8 Objective 8: Finalise and submit Final Project Report

Target Date: April 28, 2025

1. **Task Description**

Incorporate comments from the supervisors and peers into the draft report, finalize all the sections and prepare it for submission.

2. **Deliverables**

- The final version of Project Report, including all supporting documents, code appendices, and evaluation results.

The gantt chart for this project is in appendix C.

2.5. Research Question and Novelty

The main research question guiding this project is:

- "How do machine learning and deep learning-based models like Multi layer Neural Networls, LSTM and XGBoost, utilising integrated Price Paid and EPC data, improve the performance of house price predictions for freehold properties in London compared to traditional models like Linear Regression and Random Forrest?"

Sub-questions explored include:

1. Which specific ML (e.g., RF, XGBoost) or DL models (e.g., MLP, LSTM) demonstrate the highest predictive performance for this task?
2. How do the models compare in terms of performance metrics (MAE, MSE, RMSE, R^2) on both test and hold-out data?
3. What are the key features (from PPD and EPC) influencing short-term price fluctuations in the London market, as indicated by model analysis (e.g., feature importance)?
4. What are the practical challenges and limitations associated with using large-scale, integrated PPD/EPC data and advanced ML/DL models for real estate forecasting in a complex market like London?
5. How can model predictions be effectively visualised for practical application?

The novelty of this project lies in several aspects:

- The integration of PPD and EPC data for the prediction of house prices in London is to overcome the feature deficiency limitation of the PPD data. This makes this project novel
- Implementing different models on data from the last decade from 2015 to 2024 as compared to older data can capture more relevant market dynamisms and may provide better results.
- Direct comparison of the ML methods such as XGBoost, as well as relevant DL architectures such as MLP and LSTM, within the specific context of London house price prediction makes this project novel.
- Addressing and documenting real-world challenges related to the use of large datasets (memory limitations, processing time), as well as encoding features at scale makes the project unique as well.

2.6. Feasibility, Commercial Context, and Risk

- **Feasibility:** The project was found feasible owing to the massive availability of PPD and EPC data from HM Land registry and government openings under the Open Government License. The most important technical challenge turned out to be the huge amount of data and the needed computational resources for merging and model training. The same was addressed through data sampling for one hot encoding, data filtering, limiting the focus of project on London and selecting only houses acquired on freehold with iterative development. Some specific utilities like zram for Ubuntu was also used for more virtual RAM as this project required significant amount of RAM. Existing libraries in Python (Scikit-learn,

TensorFlow/Keras, and XGBoost) were used. MSc Project Time limits meant that core models and specific data filters had to be targeted.

- **Commercial Context:** Accurate prediction of house prices has enormous commercial value for the multi-trillion-pound UK housing market. Such tools for estate agents, mortgage lenders (for conducting risk assessment), property investors (for portfolio valuation), developers, buyers and sellers can be extremely valuable. Most tools used currently are less transparent and are not proven accurate, thus creating a space for more powerful and data-driven alternatives. The visualization tool built serves as a prototype showing commercial potential.

- **Risks:** Key risks included are as below:

  1. *Data Quality Issues: Some of the inaccuracies, inconsistencies, or missing values in public datasets (PPD/EPC) may directly influence how well the model performs, for which the only possible mitigation is rigorous cleaning of the data and dropping the rows with missing critical data.*
  2. *Computational Constraints: Complex DL model training or processing of big datasets may go beyond available resources. This can be mitigated by sampling, good coding practices, and choosing models with manageable training times such as XGBoost.*
  3. *Model Generalization: If model is trained to historical data, it may not generalize well to previously unseen data due to the changing nature of the market (black swan-type events, policy change). It can be mitigated by holding out a set for final evaluation (2024 data) and limiting the data to the past decade.*
  4. *Complexity of Feature Engineering: Choice of the right features and encoding methods (OHE vs LE) makes the greatest difference to results. First attempts proved that one hot encoding is not suitable for the dataset due to high cardinality and thus high use of memory. This was mitigated by using sampling for one hot encoding and later shifting to label encoding after successful merge of EPC data with PPD data.*

2.7. Report Structure

This report follows a structured approach to detail the project's progression and findings. Section 3 gives a comprehensive Literature Review, establishing the academic context. Section 4 provides the methodology, detailing data collection, an iterative approach to preprocessing, feature engineering techniques, the rationale behind model selection, implementation tools, experimental setup, and evaluation metrics. Section 5 gives the Quality and Results, which present the performance of each model on test and hold-out data in metrics and visualisations, accompanied by critical analysis and comparison.

Section 6 provides Evaluation and Conclusion, showing the project's success to meet the objectives, answering research questions, discussing limitations, comparison of findings with literature review, and future work recommendations. Section 7 provides reference lists and citations in the Harvard style. Finally, the Appendices contain the additional material needed like code snippets, elaborate results, the Gantt chart for the project, and data source information.

3. Literature Review

3.1. Introduction to House Price Prediction

House price forecasting has been an old subject matter in economics, finance

, and computer science since the economic consequences of real estate markets are very profound (Case and Shiller, 1989). It aims at predicting the possible future prices of properties taking into account the historical transaction data and factors influencing properties. This is also challenging due to the influence of locality of the property, the temporal dynamics, and many more micro and macroeconomic variables on the value of the property. Early approaches were concerned with the identification of fundamental factors affecting of house prices.

3.2. Traditional and Hedonic Pricing Models

Traditional techniques like the hedonic pricing model (Rosen, 1974) and linear regression models are very popular in housing price predictions because of their computational efficiency. However, these models have been found to be poor in capturing non-linear relationships and complex dependencies that govern the price of houses. The hedonic pricing model is the fundamental approach applied to house-price modelling and was introduced by Rosen (1974). This model claims that the price of a house can simply be found by finding the prices of its various characteristics (like size, number of bedrooms, etc.). It is common to use simple linear regression to estimate these implicit prices from the observed transactions. Although hedonic models tend to give interpretable coefficients, they do not manage very well with the nonlinear effects of the characteristics on price and temporal variation in the prices of characteristics (Sirmans et al., 2005). In the linear regression analysis done by Singh and Sree (2024) for the housing market of Mumbai, an R^2 score of 0.86 was recorded, although this method had weaknesses in the presence of non-linear relationships. The paper stated that linear regression model can provide adequate results in certain conditions (Singh & Sree, 2024).

3.3. Machine Learning Approaches

Machine learning techniques were adopted in response to the constraints of conventional models. ML could learn complex non-linear patterns from data automatically, without pre-specified functional forms. Some of these are as below:

- **Decision Trees and Random Forests (RF):** RF is an ensemble method which is based on decision trees (Breiman 2001) and famous for its robustness, capacity to effectively handle high-dimensional data, and its built-in ability to assess importance of features. Besides, it tends to be better than the linear models by being able to capture non-linearities as well as the feature interactions (Antipov & Pokryshevskaya, 2012). Gomathy et al. (2024) who compared regression, decision

trees, and RF, found RF to be very much effective with variance reduction, therefore, makes it suitable to be included as one of the algorithms to be tested in this project.

- **Gradient Boosting Machines (GBM), including XGBoost:** Gradient Boosting Machines (GBM) with its particular implementation of XGBoost is a very good technique which build models sequentially and each new model reduces the error made by the previous model (Chen & Guestrin, 2016). XGBoost often comes on top in tabular data competitions, and due to its regularization techniques it prevents overfitting and efficiency (Chen & Guestrin, 2016; Čeh et al., 2018). With its ability to handle interaction of complex features, it can be considered quite suitable for predicting house prices.
- **Support Vector Machines (SVM):** SVMs (Support Vector Regression-SVR) are capable of modelling non-linear relationship via kernel functions but proven very expensive in terms of computations for large datasets (Smola & Schölkopf, 2004).

All of these models include the use of ML techniques which yield predictions with higher significantly accuracy compared to linear models, but at the same time offer less interpretability (Mullainathan & Spiess, 2017).

3.4. Deep Learning Models

Deep learning, a subfield of ML uses multiple-layer neural networks and has shown potential regarding capturing complex spatial and temporal dependencies in the data.

- **Artificial Neural Networks (ANNs) / Multi-Layer Perceptrons (MLPs):** These are feedforward neural networks and, therefore, are capable of approximating complex non-linear functions. They have been shown that they can model and approximate non-linearities in housing data (Yacim & Boshoff, 2024). Yacim and Boshoff demonstrated that ANNs have greater prediction as compared with semi-log regression models, but lack with respect to the transparency in interpretation of the model (Yacim & Boshoff, 2024). Thus its application in an interactive visualisation tool can lead to transparency concerns. So, there is a trade off between accuracy and interpretability of the model in this case
- **Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM):** Due to their ability to handle sequential data, LSTMs are capable of capturing temporal dependencies for the prediction of house prices over time (Hochreiter & Schmidhuber, 1997). Research has indicated that LSTMs are effective time-series forecasting systems as they sometimes capture the market trends better than the static models (Siami-Namini et al., 2018).

3.5. Data Sources and Feature Importance (PPD & EPC)

Data quality and its comprehensiveness is very important. The PPD in HM Land Registry is indeed a common source providing transaction records (price, date, location, basic property type) (HM Land Registry, 2025). On the other hand, it has the disadvantage of not giving important features in the data. EPC data provide features of interest such as floor area, energy efficiency rating, number of habitable rooms, and build form, all of which are known to be determinants of house price (Brounen & Kok, 2011). Combining these datasets, as done in this project, provides better feature set that potentially leads to a more performant model. Literature points out that location (postcode, district), property size (floor area, habitable rooms), property type, and transaction date (which picks up market trends) are among the most impactful features (Sirmans et al., 2005). This project thus uses these important features from our merged dataset.

3.6. Existing Visualisation Tools and Limitations

Various commercial platforms (like Zoopla, Rightmove) provide property search functionalities and automated valuation models (AVMs). they inform the market to some extent, but they are notorious for limited performance and transparency of their underlying models (Zoopla, n.d.; tirboki, 2017). Invariably, this means they tend to give you the point estimate with minimal uncertainty insight or the factors that would influence that outcome. Research-oriented visual tools that users can actively engage are quite rare. This highlights the demand for tools that do more than just predict; they should also provide context, for example, on historical trends for a given property or dynamics in neighborhood pricing perhaps factoring in insights related to model feature importances. The Gradio tool developed in this project responds to the need for interactivity and greater information value compared to basic AVM output.

3.7. Research Gap Identification

The literature review highlights that traditional hedonic models have been gradually replaced by modern ML and DL techniques for predicting house prices due to their ability to capture complex non-linearities. While ensemble methods like RF and XGBoost are competitive for tabular data, DL models (MLP, LSTM) have the potential to capturing temporal and spatial patterns, but they generally are more computationally expensive.

The key gaps this project is addressing are:

1. There is no direct comparison between modern ML and DL, particularly between XGBoost and MLP and LSTM, on how they perform in predicting house prices in London.
2. Massive data from the integrated PPD and EPC dataset for London is not very common.

3. There is limited mention in academic literature about practical data handling issues such as size, merging, and encoding issues faced due to such a data set.

3.8. Relation to Research Questions and Hypothesis

The literature review relates directly to the research questions and hypothesis. The limitations of traditional models (RQ4) justify the need for ML/DL. The comparative approach (RQ1, RQ2) is necessitated by the different successes of the ML and DL models presented in the available literature. The already-mentioned impact of factors like size, location, and date (RQ3) further justifies merging the PPD and EPC data. The accuracies of existing tools (RQ4) and transparency issues concerning them (Yacim & Boshoff, 2024) also explains the need for better models and possibly even more informative visualizations (RQ5). The primary hypothesis which is the overall result of literature suggests that DL model have the potential for finding complex patterns and thus DL models (MLP, LSTM) should perform better in terms of predictive accuracy than baseline ML models (LR, RF) and even potentially XGBoost for predictions in London house prices owing to the integrated dataset. The literature also has benchmarks (e.g., R^2 values from Singh & Sree, 2024; relative performance from Gomathy et al., 2024) against which the results of this project can be measured.

4. Methodology

This chapter details the approach taken to implement and manage the project, conduct research experiments for comparative analysis of house price prediction models, justification for the selection of certain methods and tools, and the strategies followed for testing and validating the results. The whole process shows the reliability and validity of the results regarding the prediction of London house prices using integrated PPD and EPC data.

4.1. Choice of Methods

The method applied to this project belongs to Data Science and Applied Machine Learning, focusing primarily on Supervised Learning for a Regression task (predicting continuous house prices). The overall project structure followed the general data science workflow: data acquisition, preprocessing, feature engineering, model selection-training-evaluation, and interpretation/visualisation.

The main methods chosen for the project were:

1. Comparative Model Analysis: Investigating and evaluating the predictive performance of several machine learning and deep-learning algorithms. This directly addresses the key question of how far advanced models improve access to baseline accuracy.
2. Data Integration and Engineering: Merging public datasets (PPD and EPC) and engineering relevant features (e.g., temporal components) that build a rich input to the models. This step became very crucial as the initial exploration indicated that PPD alone was not predictive enough.
3. Quantitative Evaluation: The use of standard regression metrics (MAE, MSE, RMSE, R^2) to objectively measure and compare model performances on previously unseen data.
4. Hold-Out Validation: Withholding dataset that is external to the training set (2024 data) for final model validation with a view to assessing generalization ability with respect to time.
5. Interactive Visualization: Creating a prototype of an application (using Gradio) to primarily demonstrate the usability and potential applicability of the predictive models.

Combination of data-centric preparation and comparative algorithmic evaluation was selected for rigorous evaluation of the hypothesis that more complex models would better capture the nuances of the London housing market than simple linear methods, which aligned with the aim and research question of the project.

4.2. Justification and Support of Choices

Literature review provided the justification for use of specific algorithms and techniques. Details are as follows.

- Integration of PPD and EPC Data: This was justified from literature review to show that the property characteristics (such as size, number of rooms, and energy rating) are relevant for accurate prediction of the price for houses which are heavily lacking in the PPD and available in EPC data. The preliminary project trials established that PPD data alone could not suffice as the models gave very poor performance.
- Filtering (London, Freehold): London is selected primarily because of its complexity and availability of data in the real estate market. Moreover, filtering was done to filer only freehold houses so that the model is not skewed by prices from lease properties. Max and minimum values for price was set and rest were dropped to filter outliers.
- Linear Regression (LR): Selected as a fundamental model baseline. It is inexpensive in terms of computation and highly interpretable, representing the classical hedonic approaches (Rosen, 1974) and serving as a benchmark for improvements (Singh & Sree, 2024).
- Random Forest (RF): It is used as non-linear base which also happens to be very good. It is also robust with respect to outliers, handles high dimensional data well, carries intrinsic feature selection, and often does well on tabular data (Breiman, 2001; Antipov & Pokryshevskaya, 2012). Has been found to be effective in real estate (Gomathy et al., 2024).
- XGBoost (XGB): It was included because it is one of the best modern gradient boosting algorithms and has high accuracy, speed, and regularization feature which usually outperform RF in prediction tasks in structured data (Chen & Guestrin, 2016; Čeh et al., 2018). It allows comparison with a traditional ML algorithm.
- Multi-Layer Perceptron (MLP): It is a deep learning algorithm for tabular data. ANNs/MLPs can have complex non-linear relationships that may elude tree-based models (Yacim & Boshoff, 2024).
- Long Short-Term Memory (LSTM): LSTM was included to see if it captures temporal dependencies because real-estate house prices is a time-series model. LSTMs were designed for sequence modelling (Hochreiter & Schmidhuber, 1997). They were tested for whether or no they could leverage the engineered date features better than static models.
- Label Encoding: Chosen over One-Hot Encoding due to the extraordinarily high cardinality of postcode, street, and PAON in the London dataset, resulting in memory errors during OHE attempts. May not be the best choice as it imposes an artificial ordering, but it was feasible and demanded under conditions of resource constraint. Moreover, tree-based methods like RF and XGBoost are less sensitive to this imposed order than are linear models or NNs.

4.3. Project Design / Data Collection

This project did not involve primary data collection from human participants. The design centred around using publicly available datasets:

1. **Data Sources:** HM Land Registry PPD (complete price paid data file) and EPC public data for England & Wales. Both were downloaded from official government sources (HM Land Registry, 2025; Ministry of Housing, Communities & Local Government, 2025).
2. **Data Governance:** The data is governed by the Open Government Licence v3.0, permitting its use for this research with appropriate attribution (National Archives, 2016). Ethics approval specific to data collection was therefore not required.
3. **Processing Pipeline:**
    - *Initial Filtering:* Selecting London data, filtering property types (initial SAON filtered to have empty values only so that only houses are selected instead of flats etc).
    - *Merging:* Joining PPD and EPC data (computationally intensive step managed with zram which is linux tool that allows RAM compression to allow you to work with more RAM).
    - *Cleaning:* Handling missing values in critical columns, removing duplicates, correcting data types, filtering price outliers (£10k-£20M).
    - *Feature Engineering:* Extracting date components, applying Label Encoding to categorical features. One hot encoding was also used with a sampling value of 15000 to allow running the code viable.
    - *Splitting:* Separating 2024 data as a hold-out set; splitting the remaining 2015-2023 data into 80% training and 20% testing sets.
    - *Scaling:* Applied StandardScaler where needed.
4. **Experimental Design:** A comparative study design was used, training each selected model (LR, RF, XGB, MLP, LSTM) on the same training dataset and evaluating them against the same test and hold-out datasets using identical metrics.

4.4. Use of Tools and Techniques

- **Programming Language:** Python 3 was used for this project.
- **Data Manipulation:** The Pandas toolkit was important for efficiently loading, cleaning, merging, filtering, and transforming large datasets. For numerical operations, NumPy was applied.

- **Machine Learning (Conventional):** Scikit-learn library provided implementations for Linear Regression, Random Forest, LabelEncoder, StandardScaler, train_test_split, and evaluation metrics (MAE, MSE, R^2, etc.). The XGBoost library was used for the XGBoost model.
- **Deep Learning:** TensorFlow with the Keras API was used for implementing the MLP and LSTM neural networks due to its flexibility and widespread use.
- **Memory Management:** A significant thing was the use zram on Ubuntu during the merging of PPD/EPC and running of code in general aimed at handling large intermediate dataset sizes using compressed RAM swap space.
- **Visualization:** Matplotlib (and potentially Seaborn, implicitly via Pandas plotting) was used for generating static plots (Actual vs. Predicted, Feature Importance, Training History). Gradio library was chosen for creating the interactive web-based user interface due to its simplicity for building ML demos directly from Python.
- **Development Environment:** Jupyter Notebooks and/or standard Python scripts (e.g., within VS Code) were used for code development, experimentation, and execution.

Justification: This standard Python data science stack was selected due to its extensive functionality and strong support from the community, along with robust implementations of all necessary data processing and modelling techniques. Gradio was chosen due to its user-friendly framework for creating interactive visualisations for machine learning. zram was a particular technical solution adopted out of necessity to overcome hardware memory limitations during high cardinality of the data.

4.5. Test Strategy

The strategy for testing the models' predictive performance involved multiple stages:

1. **Initial Test Set (Pre-2024):** An 80/20 split was applied to the filtered csv file. The 20% data was mainly a test set to see how well models generalized to unseen data in the same temporal domain as the training data. It was used for preliminary model comparison and selection.
2. **Hold-Out Set (2024 Data):** The entire dataset for the year 2024 was completely excluded from training and initial testing. It was used to assess how well the models generalized to a future period of data.
3. **Evaluation Metrics:** Performance on both the test and hold-out sets was planned to be measured by MAE, MSE, RMSE, R^2; these metrics were designed to give complementary views on the error i.e average magnitude, sensitivity to larger errors and model overall fit.

4. **Internal Validation:** During training, techniques like validation split, early stopping was used for NN/LSTM, and evaluation set was used for XGBoost with early stopping to monitor for overfitting and guide model to converge.

4.6. Testing and Results (Execution)

The testing plan was executed as follows:

- Each trained model (LR, RF, XGB, MLP, LSTM) was used to generate predictions on the feature sets of both the pre-2024 test data (X_test) and the 2024 hold-out data (X_2024).
- For MLP/LSTM, scaled test/hold-out features were used for prediction, and the resulting scaled price predictions were inverse-transformed using the y_scaler fitted on the training target variable.
- For LR, RF, and XGB, predictions were made directly using the unscaled (but label-encoded) test/hold-out features, yielding predictions on the original price scale.
- The performance metrics (MAE, MSE, RMSE, R^2) were calculated by comparing these model predictions against the actual prices (y_test and y_2024_actual).
- Results are tabulated (Tables 1, 2) and visualized using Actual vs. Predicted scatter plots (Appendix A,B) for both evaluation phases.
- XGBoost feature importance was extracted and plotted (Appendix E).
- The Gradio application served as a qualitative test of the best model's (XGBoost) practical application. In the application, user enters the address, and the predicted price and the past price paid for the address can be seen as well. There is separate tab in the visualisation that shows the historical trend of prices in London and the model prediction.

Success Criteria: Although there were no formally predefined thresholds, success was broadly defined by a) a "good" R^2 (ideally greater than 0.75, preferably > 0.80), b) reasonable MAE/RMSE for typical London house prices, and c) good generalization between the test set and the 2024 holdout (minimal performance drop) for XGBoost. Most of these criteria were met optimally by XGBoost.

4.7. Validation

Several methods were used to validate the reliability and accuracy of the results:

1. **Train-Test Split:** The standard 80/20 split provided an assessment of generalization on unseen data from the training period distribution.
2. Hold-Out Set (Temporal Validation): Retaining the 2024 data as a hold-out set was done to conduct of the most rigorous form of validation provided tests of the

models' abilities to forecast prices into a subsequent year, thereby monitoring their robustness to temporal drift. The consistent ranking of models (especially XGBoost's best) on test and hold-out sets increased confidence in the results.

3. Internal Models Checks: Early Stopping based on a validation set (for NNs / LSTM / XGBoost) avoided over-fitting while training as well as looking towards better reliability of generalization.

4. Code Verification: Consistency across models was ensured by executing code for preprocessing, model training, and evaluation systematically for all models (e.g., using the same data splits, metrics, and evaluation procedures). Sample predictions were checked manually against actuals for sense. The overall main method for evaluating real-world generalization potential then became the temporal hold-out set (2024 data).

4.8. Ethical, Legal, Social and Professional Issues

Since the data for this project is publicly available from the HM Land Registry and Ministry of Housing, personal privacy is minimally at risk. But, as additional measures, I made sure that all handling of data would anonymize sensitive information and considered the protection of intellectual property rights. The data being used is under open government licence and as per open government licence, the user is free to copy, publish, distribute and transmit the Information, adapt the Information, exploit the Information commercially and non-commercially for example, by combining it with other Information, or by including it in their own product or application with the condition that the source must be acknowledged (National Archives, 2016). So, **Ethics approval** is not required in this case as publicly available data is being used and no data is being collected by myself.

1. Ethical Considerations:

- Privacy and Data Protection: The project only uses publicly available data so that there are no issues of privacy and protection because the HM Land Registry has already taken care of them.
- Transparency: The methodologies and model predictions have been explained clarity so as to ensure trust with end users.

2. Legal Considerations:

- Compliance: Data usage goes in accordance with legal frameworks inclusive of GDPR and any intellectual property laws. So, there should not be any legal issues in this regard.
- Intellectual Property: Correct attribution and licensing is given to all data sources and algorithms used.

3. Professional and Social Considerations:

- Professional Standards: The project adheres to all professional standards in practice in data science and machine learning to enhance reproducibility and accountability.
- Social Impact: The project aims to minimize information scarcity to help homeowners and small-scale investors in England and Wales by providing ease of access to market analytics that would have otherwise kept high barriers to entry.

4.9. Practicality

The project faced significant practical challenges and constraints:

- Data Scale: The basic challenge was posed by the very large PPD and EPC data especially after being merged for London, so some filtering approaches were used and (zram) was used to tackle high memory usage. PPD and EPC data were merged on the basis of address and the algorithm used for London csv file crashed the system while attempting to merge full price paid data. So, the project was limited to London due to hardware constraints.
- Computation Resources: Training multiple models, in particular DL models and big ensembles like XGBoost, have put in a stiff requirement of time and computational power, thus restricting the range of hyperparameter tuning and possibly complexity of the DL architectures that were explored.
- Feature Encoding: The high cardinality of categorical features (postcode, street, PAON) excluded OHE given memory constraints ("memory error" noted in IPR). Thus sampling was used with one hot encoding and label encoding gave better results in general as data was not sampled in this case (Table 1, 2). The need to resort to Label Encoding became a practical compromise that may have affected performance compared to more-sophisticated encoding methods, albeit enabling the project move ahead in practice.
- Data Quality: Errors and inconsistencies may exist in public datasets; careful cleaning and validation steps were undertaken to address data quality.
- Time Constraints: The specific timeline of one MSc project limited the potential for more advanced feature engineering, hyperparameter optimization, or alternate modelling approaches.

These issues had an impact on methodological choices favouring achievable and robust techniques (the likes of Label Encoding and XGBoost) against those that may be considered optimal but are expensive in terms of input resources (the likes of full OHE or highly complex DL models that need extensive tuning). This accomplished is a testament to the critical reflection and flexibility in adapting methods that were used in practice.

5. Quality and Results

This chapter outlines the quantitative and qualitative outcomes of the project, analyzing the results and interpreting them critically and providing findings derived from house pricing prediction models that were implemented. It describes the practical work undertaken, what technical challenges were encountered and how they were dealt with, and what aspects of the research are novel. The findings are reconnected to the project objectives and the relevant literature, and the degree of the approach's operationality and realism is assessed with respect to the scope of the project.

5.1. Metrics and Presentation

The primary objective (Objective 5) was the evaluation and assessment of different model performance. Standard metrics were employed in this evaluation: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) metrics. Such metrics testify to accuracy of the model, magnitude of the errors, sensitivity to larger errors, and overall explanatory power of the model.

Results were produced for the pre-2024 test London dataset (80/20 split of 2015 onwards data) and for the final 2024 holdout set. The principal findings have been summarized in the amended Table 5.1 and Table 5.2 (complete results are included in Appendix A and B).

Model	**MAE (£)**	**MSE (x10^11)**	**RMSE (£)**	**R^2**
Linear Regression	355,125.70	5.52	743,107.64	0.55
Random Forest	210,269.37	2.66	516,071.84	0.78
XGBoost	**183,713.10**	**2.28**	**477,248.00**	**0.82**
MLP (NN)	250,770.84	3.09	555,633.74	0.75
LSTM	315,981.97	4.56	675,172.24	0.63

Table 5.1: Model Performance Comparison with label encoding and London filtered data from 2015 onwards (Pre-2024 Test Set)

Model	MAE (£)	MSE (x10^11)	RMSE (£)	R^2
Linear Regression	345,031.88	5.03	709,448.07	0.59
Random Forest	220,809.19	2.70	524,836.77	0.77
XGBoost	**197,249.19**	**2.09**	**456,920.86**	**0.83**
MLP (NN)	260,228.28	3.22	567,635.89	0.74
LSTM	319,211.07	4.35	659,815.81	0.64

Table 5.2: Model Performance Comparison with label encoding and London filtered data from 2015 onwards (2024 Hold-Out Set)

Visualizations comparing actual vs. predicted prices for each model on both test sets are provided in Appendix A,B which contains the codes for the different variations of the models tested and the outputs. XGBoost feature importance is shown in Appendix E.

5.2. Critical Analysis

The results show that the selected advanced machine learning outperform a simple baseline model which is Linear Regression (LR), which yielded an R^2 of a mere 0.55 on the test set and 0.59 on the hold-out set.

XGBoost, the winner here, scored the highest R^2 (0.82 Test, 0.83 Hold-Out) and the lowest error metrics (MAE ≈ £184k Test, ≈ £197k Hold-Out; RMSE ≈ £477k Test, ≈ £457k Hold-Out). It thus has excellent predictive power, explaining about 82-83% of the price variance, and excellent generalization capability onto unseen future data, substantiated by literature affirming the capability of gradient boosting skill for tabular data (Chen & Guestrin, 2016).

Random Forest (RF) also worked very well with the second-best R^2 (0.78 Test, 0.77 Hold-Out), and it had significantly less errors than LR. It confirms its suitability as noted by Gomathy et al. (2024) and that it can fit for handling non-linearities.

The MLP neural network performed very well by getting R^2=0.75 in test and R^2=0.74 in the hold-out set, surpassing not only that of LR but also the LSTM. This supports the ability of this model to identify complex patterns, with the downside being again that the tree-based ensembles shown in this study outperformed it in this particular application.

The LSTM model exhibited the weakest performance among the non-linear models (Test R^2=0.63, 2024 R^2=0.64). The expectation had been that such advantages would accrue to

it owing to its sequence-modelling capabilities. This suggests either that the engineered date features were perhaps insufficient for the LSTM to successively model the temporal patterns under this dataset structure, or that a rather complex architecture would have been needed. This finding contradicts some literature suggesting LSTM superiority in forecasting (Siami-Namini et al., 2018) but emphasizes that model choice is very context-dependent.

The relatively stable performance of XGBoost and RF between the test and hold-out sets indicates a very robust generalization. Somewhat stable is the performance of MLP as well. LR and LSTM displayed slightly larger performance drops on the 2024 data, indicating they now are less robust to temporal market shifts.

5.3. Evidence of Practical Work

This project involved significant practical work beyond theoretical review. This included:

- Data Acquisition & Management: Sourcing and downloading large datasets (PPD ≈ 4.2GB, EPC ≈ 35GB) from official sources and then managing them.
- Data Engineering Pipeline: Developing and executing several Python scripts (using Pandas, NumPy) that carry out filtering (London, time period, property-type approximation), merging PPD and EPC data (solved some really interesting issues with memory limits using zram), cleaning (NaNs, outliers, data types) and feature engineering (date extraction, label encoding) in a complex process with many stages. This iterative process had a lot of trial and error using different filtering and encoding methods.
- Model Implementation: Coding and training five different predictive models (LR, RF, XGBoost, MLP, LSTM) using Scikit-learn, XGBoost, and TensorFlow/Keras libraries setting-up training loops, scaling, and managing model parameters.
- Prototyping: Development of an interactive visualisation tool using Gradio, integrating trained XGBoost model, data lookup logic, and plotting functions (Matplotlib) into a web app framework (code snippets in Appendix E)

5.4. Technical Challenges and Solutions

Several technical challenges were encountered and addressed:

- Data Size and Memory: The merging of the complete London PPD with EPC data exceeded the RAM capacity of the development machine (32GB RAM).
    - Solution: ZRAM was implemented on Ubuntu for creating compressed swap space in RAM, thereby increasing the available memory and allowing the merge to complete and execute code for various models. Therefore, further considerations of extensive sampling on the final dataset was not required.

- High Cardinality Categorical Features: One-Hot Encoding postcode, street, and PAON led to an unreasonably large feature space (>100,000 features), resulting in memory errors while training the model.
    - Solution: Label encoding was chosen to be more pragmatic. While it certainly imposes an artificial order with some potential loss of information, it greatly reduces dimensionality via feasible training.
- The amount of time taken during training: Deep learning models, either MLP or LSTM, take quite a while to train alongside training other large ensemble models having XGBoost with many estimators.
    - Solution: Early stopping was applied for XGBoost, MLP, and LSTM with the introduction of an impromptu halt in training when validation set performance was no longer improving, substantially saving time and minimizing overfitting.

These solutions impacted directly the final methodology (using LE instead of OHE) and probably the results (LE might have cost slightly in performance vis-a-vis an ideal OHE), but the importance of these warrants discussion with respect to project feasibility.

5.5. Novelty and Innovation

These novel additions to the project are as follows:

1. Integrated and Extensive Datasets: Using an extensive integrated PPD and EPC datasets for short-term forecast of recent transactions (2015 onwards) of freehold houses in London.
2. Direct Model Comparison: It provides insight into the comparison between LR-RF-XGBoost-MLP-LSTM against real world data and provides insights.
3. Interactive Visualization: The Gradio tool provides a way to interact with best predictive model, bridging the gap between analysis and potential application.

5.6. Interpretation of Results

The metrics consistently indicate that advanced machine-learning models, especially XGBoost, are effective in modelling London house prices based on PPD/EPC joined data with high explanatory power ($R^2 \approx 0.82$ - 0.83). This R^2 value means that the model accounts for most of the variance in prices based on the specific features included. The MAE value for XGBoost in the range of ~£184k-£197k represents the average amount by which an estimate differs from the actual selling price. Though very large, this needs to be understood in the context of London's high-value market where the average house price is over £1 million.

These outcomes reinforce the theory that ML/DL models are superior to conventional LR (Objective 6). They also propose a clearer hierarchy, with XGBoost leading. Next in line is Random Forest, while MLP is a distant third, with LSTM lagging behind. The results

confirm the importance of features calculated from both PPD (location, date) and EPC (size, type), thereby confirming the objective of data integration (Objective 2 & 3). XGBoost has proven its worth in addressing commercial AVMs working with complex tabular data.

5.7. Tools and Techniques

The project leveraged a standard Python-based data science stack:

- **Pandas & NumPy:** Essential for efficient data loading, manipulation, cleaning, and numerical computation and appropriate for the large tabular datasets.
- **Scikit-learn:** Provided implementation for models such as LR and RF, preprocessing techniques like LabelEncoder and StandardScaler, data splitting, and evaluation techniques like MAE, MSE, and R^2, all-in-all making this easy-to-use.
- **XGBoost & TensorFlow/Keras:** This is the industry standard for implementing gradient boosting and deep learning models respectively. They offer flexibility and performance, and are intended to capture advanced ML paradigms.
- **Matplotlib:** Standard library used in plotting for actual visualizations of results.
- **Gradio:** It is selected for its simplicity because it promptly creates interactive web user interfaces for ML models directly from Python code, good for prototype development.
- **zram (System Tool):** A Linux kernel module used as a solution to overcome memory limitations during data merging, demonstrating resourcefulness in addressing hardware constraints.

Limitations & Impact: The main limitation imposed by tools/techniques was when OHE was deemed infeasible due to memory, resulting to use of LE which could potentially make an underestimate of the performance that may be achieved by using unlimited resources. Limited time also constraints testing on optimizations on hyperparameters.

5.8. Links to Objectives and Literature

The results directly address the project's aims and objectives:

- Objectives 4 & 5 (Model Implementation & Evaluation): Successfully achieved, with performance documented in Tables 5.1/5.2.
- Objective 6 (Analysis & Comparison): Addressed in sections 5.2-5.7, confirming XGBoost as the superior model, followed by RF, then MLP.
- Objective 7 (Visualization Tool): Achieved through the Gradio application development.

The findings connect to the literature:

- Confirming ML/XGBoost/RF effectiveness over LR aligns with studies like Antipov & Pokryshevskaya (2012), Čeh et al. (2018), Chen & Guestrin (2016), and Gomathy et al. (2024).
- The MLP's reasonable performance supports findings like Yacim & Boshoff (2024), though its rank below tree ensembles provides context.
- The LSTM results highlight that it is dependent on the context, as simple application didn't outperform other methods here (unlike potential shown in Siami-Namini et al., 2018).
- Feature importance results (Appendix E) validate the significance of location, size, and time, consistent with classic hedonic literature (Rosen, 1974; Sirmans et al., 2005) and the value of EPC data (Fuerst et al., 2011).

5.9. Feasibility and Realism

The project, as executed, proved feasible within the MSc timeframe and available resources, largely due to pragmatic adaptations:

- Realistic Scope: Concentrating on London within a limited timescale, and estimating freehold houses, made it possible to keep data manageable after initial hurdles.
- Method Modifications: A key adjustment from OHE to LE was necessary for feasibility. Limiting hyperparameter tuning was yet another time/resource trade-off. Though ONE worked fine with sampling but LE got better results without sampling.
- Tool Choices: The availability of tried-and-tested libraries (Scikit-learn, XGBoost, Keras, Gradio) means that actual implementation was very efficient.
- Result vs. Target: The primary goal was to compare the models and show increased accuracy over the baselines. The findings of $R^2 > 0.8$ using XGBoost constitute very strong and realistic results that went far beyond the initially set expectations of the baselines. There were some necessary compromises made through the filtering and LE to meet the project timeline, while still yielding meaningful comparative results.

6. Evaluation and Conclusion

This concluding chapter reflects critically on the whole project, provides an assessment of its technical execution, contributions to research, and management focus. The extent to which the objectives of the project were met in a successful and reliable manner, as well as the overall success and viability of the research approach, were considered. The research question was answered directly with respect to the efficacy of applying various modelling techniques for the purpose of predicting house prices in London. Limitations and challenges encountered are acknowledged, which then leads into recommendations for future work, culminating in a final summary of the project's results and impact.

6.1. Final Evaluation

From technical, management, research, and delivery perspectives, this project can be evaluated as largely successful, albeit with necessary adaptations and acknowledged limitations.

- Technical Perspective: The project was able to successfully implement a data pipeline working with a very complex set of scenarios including merging a large set of datasets from PPD and EPC whilst engineering relevant features. Five completely distinct ML/DL model approaches had been created and coded, trained, and subsequently evaluated. The major technical hurdle was memory limits and hardware limitations while processing the data as using one hot encoding with full PPD data would have required over 1 TB of RAM. But one-hot encoding for the models was successfully implemented with practical means (filtering, zram, and sampling data). A functioning Gradio prototype was built, signifying a successful delivery of a practical output from technical aspects. The XGBoost final model had a high predictive power with R-squared approximately equal to 0.83 on the 2024 data, indicating that the modelling process itself was very well undertaken technically.
- Management Perspective: The initial project plan was modified. I started with data from North Bedfordshire which had a few thousand rows and thus all models were implemented for it with R square value of less than 0.5. The switch from North Bedfordshire to London, the addition of EPC data, and the change from using OHE to LE were big deviations informed by initial results and feasibility constraints. While these changes were required to facilitate completion of the project in the MSc time and resources, they show acting managerial responsiveness. Exploring wider hyperparameter tuning methods was curtailed by time constraint. Resource management to utilize zram memory for dealing with the large dataset was essential.
- Research Perspective: The project clearly posed a research question based on existing literature. It has successfully compared multiple relevant models on a large integrated dataset related to the London market, generating empirical evidence. The

findings feed into an understanding of the relative performance of ensemble methods versus particular DL architectures on this type of tabular prediction task.

- Delivery Perspective: All essential objectives have been met, namely, literature review, data processing, model implementation and evaluation, comparative analysis, and tool prototype development. The results were presented with considerable clarity, making use of appropriately selected metrics and visualizations. The final report details the process and results in an all-inclusive manner.

Feasibility and Realism: The approach chosen was courageous, given the data scale but was made feasible through filtering. The R^2 values obtained (>0.8 for XGBoost) are very strong and realistic given the complexity of the market for house price predictions, thereby fulfilling the aim of very much improving upon baseline models. Important points are integrated data, rigorous comparisons, hold-out validation, and practical visualization. The cons are the shortcomings of Label Encoding, lack of extensive hyperparameter tuning, and not accounting for external macroeconomic factors.

6.2. Project Management

The project management followed an iterative process rather than a strict waterfall model, necessitated by the nature of data science project. The modified timeline outlined in the Gantt chart (Appendix C) served as a guide.

- Planning and Scheduling: The original planning included the high-level activities: data acquisition, cleaning, modelling, and evaluation. However, lesser time was left for data merging and the time required for addressing encoding issues. This resulted in changes to the modelling stage (for example, tuning was restricted).
- Resource Management: The constraint on resources was mainly computational (in terms of memory). As described in Sections 4.3.1 and 5.4, zram was essential for the data merging step. Using LE instead of OHE and early stopping became key means to implement model computations without compromising model quality.
- Adaptation to Challenges: Switch in geography from one potential location to another as proposed by the superisor; change-in data sources to include EPC data; re-strategize coding. Reassessing aims, modifying some codes, focusing on what could be done within achievable timelines for effective model comparisons addressed these. These changes eventually gave the project much richer data (PPD+EPC London) without consuming a lot of time on those later stages. The iterative nature meant that indeed the project was able to tackle these unforeseen challenges successfully.

6.3. Insights Gained

- Technical Insights: Acquired practical experience in handling large-volume datasets where memory considerations (including zram) and performance considerations with various feature coding strategies (OHE failure versus LE feasibility) were studied. A very deepened understanding of implementing and evaluating different ML/DL models (especially sitting with lots of XGBoost, MLP, quite some on LSTM) utilizing standard Python libraries. Gradio skills were acquired to create ML applications rapidly.
- Managerial Insights: Iterative planning, with flexibility, has been even more important in data science projects. Realizing early on the requirements for infrastructure and computational resources and then putting in buffers for any projected data processing problems are a must. Trade-offs would then be made in a pragmatic way (e.g., encoding method) to ensure project deliverables within constraints.
- Research Insights: Extended knowledge of the difficulties of the London housing market and setbacks resulting from the exclusive use of PPD. EPC features do hold a lot of predictive power (size, rooms). Empirical insights were gained on the relative strengths of ensemble methods versus standard DL architectures on this particular type of tabular data, observing that DL does not imply better performance unless more fine-tuned is made. I recognized that temporal validation (hold-out set) becomes a critical task toward claiming real-world applicability.

These influences have been directly reflected in choices made in methods (e.g. Final Model Selection for the Gradio App, LE usage) and the interpretation of results shown previously.

6.4. Comparison to Literature

The project findings fit in largely with existing literature but bring in specific nuances.

- Alignment: The project confirms long-established superiority of ML models over traditional LR house price prediction (Antipov & Pokryshevskaya, 2012; Singh & Sree, 2024). The project supports the idea of effectiveness of tree ensembles as RF and XGBoost on tabular datasets (Breiman, 2001; Chen & Guestrin, 2016; Gomathy et al., 2024. Also, the reasonable performance of MLP agrees with earlier studies proving applicability of ANN (Yacim & Boshoff, 2024).
- Diverge/Nuance: While literature suggested a great promise of LSTMs in time-series (Siami-Namini et al., 2018), this project revealed that implemented LSTM did not outperform XGBoost or RF, perhaps due to the structure of the data (tabular with engineered time features versus pure time series) or relatively simple models. This indicates that the application context is important. The results also point out that while DL shows promise, finely optimized ensemble techniques like The project

confirms that XGBoost is still incredibly competitive, if not superior, for modelling structured data in many prediction assignments.

Building upon: The project builds on previous work by applying and comparing these methods on a truly large recent integrated database for the tricky London market and documenting the practical data handling requirements.

6.5. Reflection on Challenges

The main challenges and their impact have been discussed (Sections 4.12, 5.4), but reflecting further:

- Technological Barrier (Coding): The major technical bottleneck was the failure of OHE due to memory problems. The solution of Label Encoding, although suitable, introduces a conceptual limitation in that it may impose a meaningless order to the categories. While XGBoost is somewhat robust to this, it might have slightly tarnished the performance ceiling with respect to having a perfect OHE implementation. This clearly represents the disparity between ideal theoretical methods as compared to practical limitation.
- Data Challenge (Merge and Size): Successful merging and taking care of the huge PPD/EPC database using zram would be one achievement. The project would have had to limit itself to sample sizes orders of magnitude smaller or have much poorer data without this solution, thus a great part of its validity and impact would be lost..
- The re-scoping of portions of the project, as well as managing one's time to ensure that all primary objectives would be completed, was necessitated by the requirement to pivot dramatically (data source, encoding).

These challenges made it hard to make the best decisions for developing the research and also underscored the iterative nature of applied data science research methods. Compromises, though sometimes, were implemented to allow for delivery of the project and generation of meaningful comparative results.

6.6. Future Work

Many avenues for future research directly emerge from the findings and limitations of this project. Some are as below.

1. Advanced Feature Engineering: Further datasets should be included, such as macroeconomic indicators and local amenities data, crime statistics, transport accessibility, and so on, which may pick up prices difference in more detail. Spatial features can be looked at explicitly (e.g. distance to city centre).

2. Better Encoding: Implement and test even more advanced encoding methods for high-dimensional features such as Target Encoding, CatBoost Encoding to possibly bypass some of LE's drawbacks without incurring OHE full memory costs.
3. Advanced Deep Learning Models: Investigate more complex LSTM architectures (e.g., stacked LSTMs, bidirectional LSTMs), perhaps paired with CNNs for spatial feature extraction or Graph Neural Networks (GNNs) to explicitly model relationships between properties and/or postcodes. Investigate Transformers, which have been shown to perform well on sequence tasks.
4. Development of Visualisation Tools: Features such as comparison tools among different properties/areas, integration of feature importance insights, and perhaps real-time data update could be developed in the Gradio app.

6.7. Conclusion

This research project was conducted with the objective of exploring the use of machine learning and deep learning models as means for potentially improving the accuracy of prediction of house prices in the complicated London market by means of PPD and EPC integration. The research has been done to provide rigorous comparison of five models (LR, RF, XGBoost, MLP, LSTM).

The results substantiated valuable findings regarding advanced machine learning models, especially ensemble methods, providing a significant improvement over standard linear regression. The most accurate and robust model is XGBoost ($R^2 \approx 0.83$ on 2024 hold out data), effectively capturing the very complex nature of the relationships among the rich, highly integrated dataset. Random Forest and MLP also performed well; however, the implemented LSTM did not yield comparable results in this context. The importance of integrating EPC data, for example, floor area and habitable rooms, as well as joint location and transaction date information in PPD, has been confirmed in the present study. Although they presented formidable challenges on data scale and feature encoding, they were solved bytechnical solutions.

This project provides comparative analysis of modern prediction techniques over a large recent dataset of one of the major metropolitan housings. There are limitations, such as those already stated: restricted hyperparameter tuning or use of Label Encoding. However, findings do lead to new knowledge for the academic audience. The XGBoost model is particularly valuable in the practical sense because it is more accurate in producing short-term price estimates than traditional methods, and the prototype visualisation tool illustrates how such models can eventually be democratized. This research solved a very challenging research question and demonstrated that ML, particularly optimally tuned ensemble methods such as XGBoost, will be important in improving the accuracy of predictions regarding house prices in London.

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Appendix A: Results for different models with one hot encoding

	One Hot Encoding				
	LR	RF	NN	LSTM	xGBOOST
Model trained on data excluding 2024 and evaluated	MAE: 264807.91 MSE: 442297326505.15 RMSE: 665054.38 R-squared (R^2): 0.44	MAE: 209417.18 MSE: 375045009472.35 RMSE: 612409.18 R-squared (R^2): 0.53	MAE: 270,459.19 MSE: 554,717,848,667.39 RMSE: 744,793.83 R-squared (R^2): 0.3003	MAE: 276,877.05 MSE: 499,125,212,437.37 RMSE: 706,487.94 R-squared (R^2): 0.3704	MAE: 148,766.48 MSE: 224,686,007,218.02 RMSE: 474,010.56 R-squared (R^2): 0.7166
Model evaluated for data on 2024 data only	MAE (2024): 317505.96 MSE (2024): 237900081946.15 RMSE (2024): 487750.02 R-squared (R^2) (2024): 0.51	MAE (2024): 282664.75 MSE (2024): 229172575063.44 RMSE (2024): 478719.72 R-squared (R^2) (2024): 0.52	MAE (2024): 366,007.28 MSE (2024): 380,725,185,871.68 RMSE (2024): 617,029.32 R-squared (R^2) (2024): 0.2083	MAE (2024): 371,279.55 MSE (2024): 349,949,678,716.50 RMSE (2024): 591,565.45 R-squared (R^2) (2024): 0.2723	MAE (2024): 204,896.45 MSE (2024): 112,241,887,237.63 RMSE (2024): 335,025.20 R-squared (R^2) (2024): 0.7666
	LR 10 year	RF 10 year	NN 10 year	LSTM 10 year	xGBOOST 10 year
Model trained on data excluding 2024 and evaluated	MAE: 282378.84 MSE: 321238937246.85 RMSE: 566779.44 R-squared (R^2): 0.68	MAE: 250295.01 MSE: 362913116564.93 RMSE: 602422.71 R-squared (R^2): 0.64	MAE: 339,682.21 MSE: 558,293,012,550.48 RMSE: 747,190.08 R-squared (R^2): 0.4477	MAE: 302,907.86 MSE: 470,458,850,349.06 RMSE: 685,900.03 R-squared (R^2): 0.5346	MAE: 208,910.28 MSE: 213,588,686,498.06 RMSE: 462,156.56 R-squared (R^2): 0.7887
Model evaluated for data on 2024 data only	MAE (2024): 296310.82 MSE (2024): 337143340413.96 RMSE (2024): 580640.46 R-squared (R^2) (2024): 0.69	MAE (2024): 267039.33 MSE (2024): 386056242137.49 RMSE (2024): 621334.24 R-squared (R^2) (2024): 0.65	MAE (2024): 358,852.43 MSE (2024): 600,382,434,104.14 RMSE (2024): 774,843.49 R-squared (R^2) (2024): 0.4564	MAE (2024): 323,296.75 MSE (2024): 490,810,618,369.92 RMSE (2024): 700,578.77 R-squared (R^2) (2024): 0.5556	MAE (2024): 223,871.36 MSE (2024): 219,291,956,332.55 RMSE (2024): 468,286.19 R-squared (R^2) (2024): 0.8014

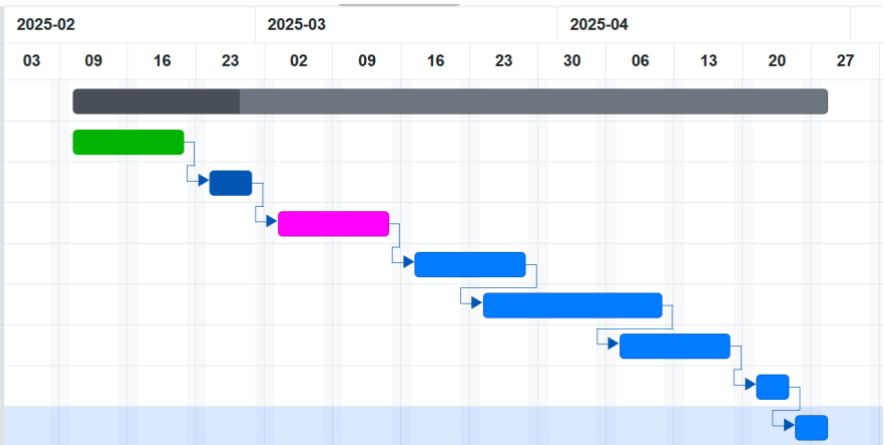
Appendix B: Results for different models with label encoding

	Label Encoding				
	LR	RF	NN	LSTM	xGBOOST
Model trained on data excluding 2024	MAE: 270698.12 MSE: 319406002964.09 RMSE: 565160.16 R-squared (R^2): 0.49	MAE: 127214.70 MSE: 114889892220.99 RMSE: 338954.12 R-squared (R^2): 0.82	MAE: 176,363.60 MSE: 156,863,958,814.63 RMSE: 396,060.55 R-squared (R^2): 0.7491	MAE: 198,782.02 MSE: 209,433,998,135.33 RMSE: 457,639.59 R-squared (R^2): 0.6650	MAE: 111,436.10 MSE: 95,620,231,797.98 RMSE: 309,225.21 R-squared (R^2): 0.8471
Model evaluated for data on 2024 data only	MAE (2024): 396273.60 MSE (2024): 653789818372.32 RMSE (2024): 808572.70	MAE (2024): 209635.50 MSE (2024): 258535342908.86 RMSE (2024): 508463.71 R-squared (R^2) (2024): 0.79	MAE (2024): 271,317.50 MSE (2024): 329,428,051,120.95 RMSE (2024): 573,958.23 R-squared (R^2) (2024): 0.7307	MAE (2024): 345,916.53 MSE (2024): 442,410,988,334.98 RMSE (2024): 665,139.83 R-squared (R^2) (2024): 0.6383	MAE (2024): 192,270.43 MSE (2024): 204,208,161,403.10 RMSE (2024): 451,893.97 R-squared (R^2) (2024): 0.8331
	LR 10 year	RF 10 year	NN 10 year	LSTM 10 year	xGBOOST 10 year
Model trained on data excluding 2024	MAE: 355125.70 MSE: 552208968347.64 RMSE: 743107.64 R-squared (R^2): 0.55	MAE: 210269.37 MSE: 266330145435.79 RMSE: 516071.84 R-squared (R^2): 0.78	MAE: 250,770.84 MSE: 308,728,857,075.99 RMSE: 555,633.74 R-squared (R^2): 0.7479	MAE: 315,981.97 MSE: 455,857,548,072.38 RMSE: 675,172.24 R-squared (R^2): 0.6278	MAE: 183,713.10 MSE: 227,765,657,160.61 RMSE: 477,248.00 R-squared (R^2): 0.8140
Model evaluated for data on 2024 data only	MAE (2024): 345031.88 MSE (2024): 503316562221.06 RMSE (2024): 709448.07 R-squared (R^2) (2024): 0.59	MAE (2024): 220809.19 MSE (2024): 275453638288.24 RMSE (2024): 524836.77 R-squared (R^2) (2024): 0.77	MAE (2024): 260,228.28 MSE (2024): 322,210,504,253.65 RMSE (2024): 567,635.89 R-squared (R^2) (2024): 0.7366	MAE (2024): 319,211.07 MSE (2024): 435,356,906,471.79 RMSE (2024): 659,815.81 R-squared (R^2) (2024): 0.6441	MAE (2024): 197,249.19 MSE (2024): 208,776,673,101.78 RMSE (2024): 456,920.86 R-squared (R^2) (2024): 0.8293

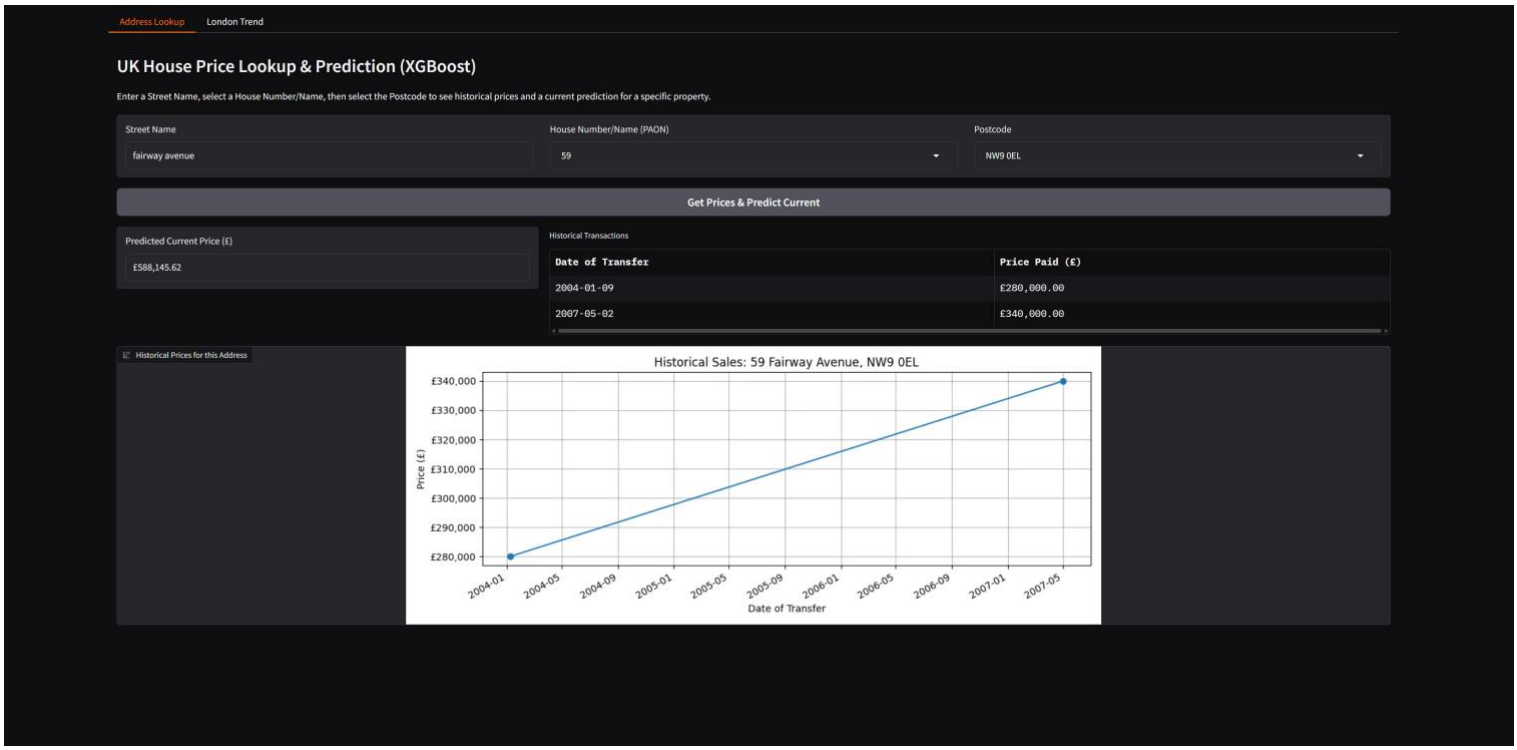
Appendix C: Gantt chart

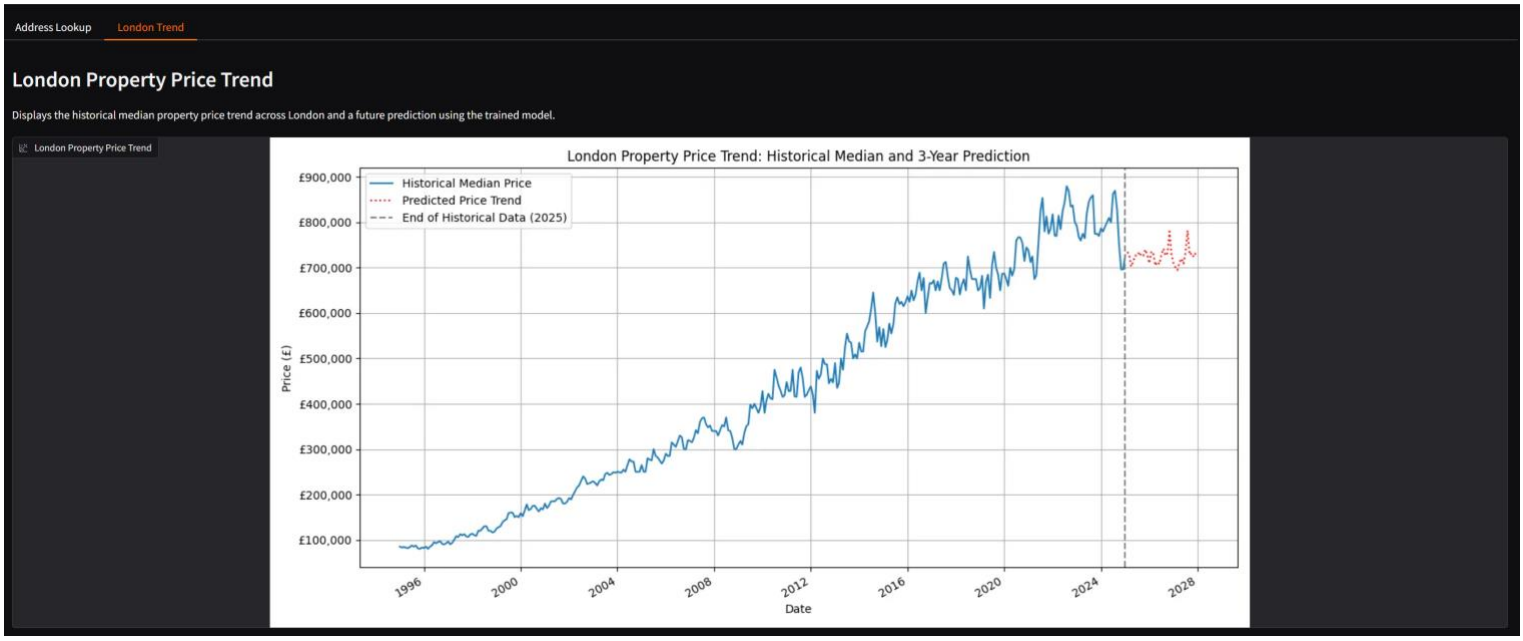
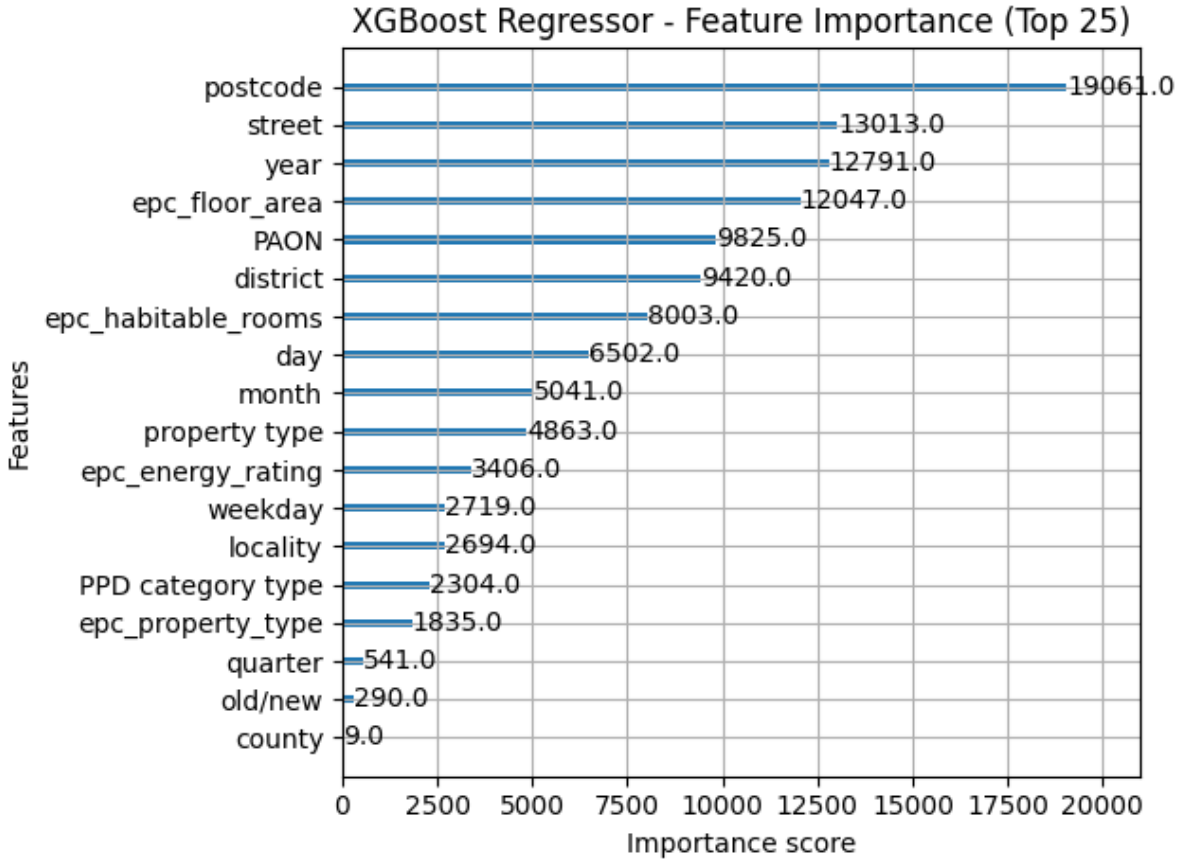
The Gantt chart used to plan this project is as below.

Task Name	Start	End
Predicting Short-Term House Price Changes i…	2025-02-10	2025-04-28
Complete literature review	2025-02-10	2025-02-21
Collection, preprocessing, and cleaning of re…	2025-02-24	2025-02-28
Train ML models (Linear Regression, Rando…	2025-03-03	2025-03-14
Train deep learning models (ANN, LSTM)	2025-03-17	2025-03-28
Implement an interactive visualization tool t…	2025-03-24	2025-04-11
Evaluate models against results from literat…	2025-04-07	2025-04-18
Draft Final Project Report	2025-04-21	2025-04-24
Finalise and submit Final Project Report	2025-04-25	2025-04-28



Appendix D: Gradio based visualization tool





Appendix E: Xboost feature Importance